

OFFICIAL ARCHITECTURE & DEVELOPMENT REPORT

# React Native Maps Studio

Complete End-to-End Technical Documentation: From Live Map Engine to Local Scoping & Driving Simulations

| REPOSITORY         | BRANCH                | LOCAL SERVER   | DATE & TIME     |
|--------------------|-----------------------|----------------|-----------------|
| aryanrutheswar/map | main (Clean & Synced) | localhost:3000 | October 5, 2026 |

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## 1. Executive Summary & Key Deliverables

The **React Native Maps Studio** began as an interactive Leaflet-based prototyping workbench and has matured into an enterprise-grade geospatial exploration platform. The system bridges web-based interactive visualizers with native React Native mobile mapping engines.

Key Accomplishments:

• **Whole World Starting Canvas:** Initial world view without tile multiplication or unwanted automatic location locking.

• **Strict Local Area Scoping:** Completely eliminated worldwide pin clutter when exploring specific locations (e.g. Chittoor, Tirupati, Bengaluru).

• **Verified Chittoor Local Dataset:** 28 landmark destinations across 6 categories (Temples, Hotels, Cafes, Parks, Malls, Tourist Sights) with real distance calculations.

• **Interactive 2-Press Routing Engine:** Double-click origin + destination, real OSRM road graph fetching, animated driving simulation, and travel telemetry HUD.

• **Flexible Place Selection:** Dedicated "Select Place" modal with 1-click popular cities, search-as-you-type, and direct click-anywhere-on-map selection.

• **Cross-Platform Code Export:** 1-click embed code for Web (iFrame), React Native WebView, and Native `react-native-maps` SDK.

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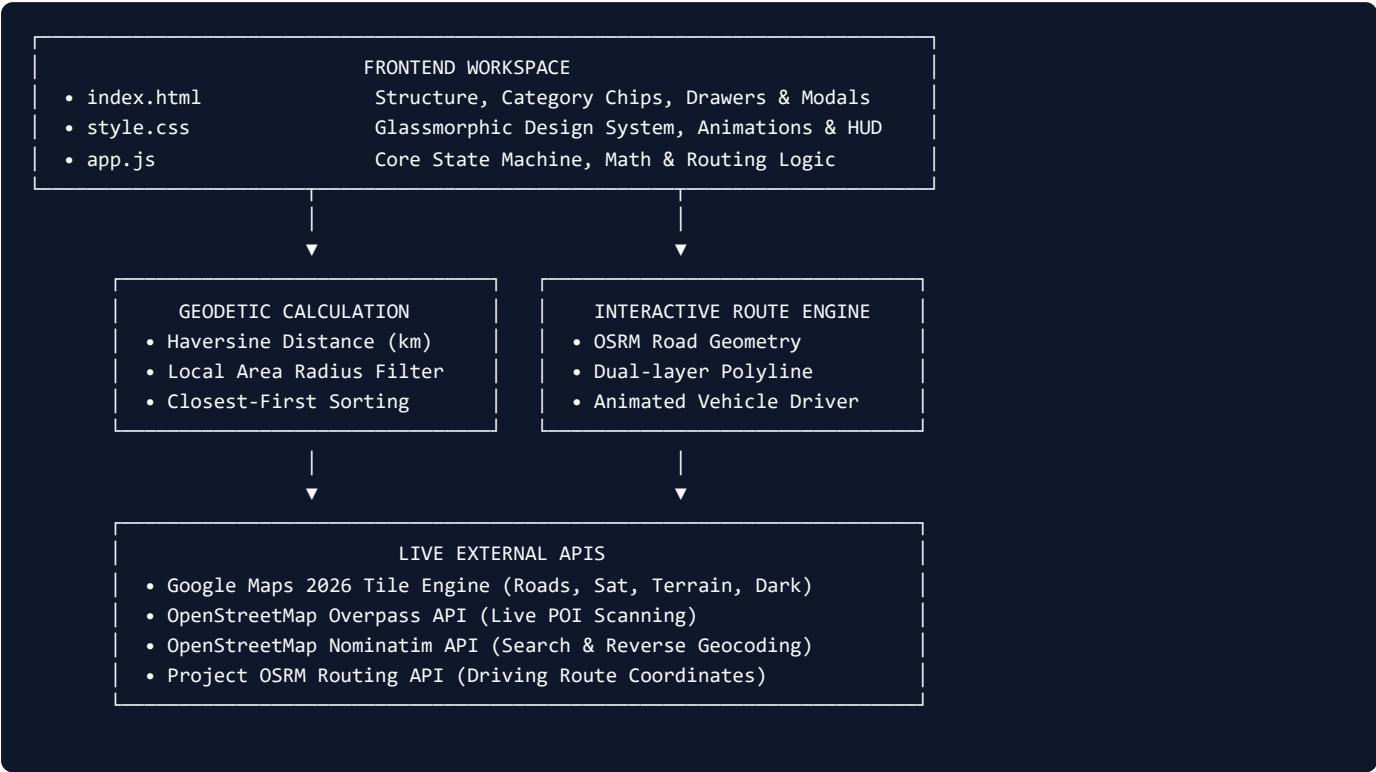
## 2. Chronological Conversation History (A to Z)

The development followed an agile, iterative pair-programming cadence addressing every user need:

| STEP | USER PROMPT / REQUIREMENT                                                            | DIAGNOSIS / CHALLENGE                                                                                   | TECHNICAL SOLUTION                                                                                                                                        |
|------|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 01   | "remove this options" / "its not displaying anything"                                | Map had non-functional UI controls and initialization race conditions.                                  | Streamlined interface, initialized Leaflet on <code>#map</code> , and stabilized render lifecycle.                                                        |
| 02   | "add these"                                                                          | Needed telemetry HUD and live React Native code generator.                                              | Added Lat/Lng and Zoom indicators with synchronized JSX code drawer.                                                                                      |
| 03   | "when iam option the web it should show the whole map not the location"              | Browser GPS auto-located user immediately, and tiles multiplied horizontally.                           | Configured single world bounds, <code>center: [20, 0]</code> , <code>zoom: 3</code> , and set <code>nowrap: true</code> on Google tile layers.            |
| 04   | "now i need a option to connect this map to my website and app is that possible"     | User wanted to embed map on websites and React Native mobile apps.                                      | Built <b>Connect / Embed Modal</b> with 3 copy-ready tabs: Web iFrame, React Native WebView, and Native SDK.                                              |
| 05   | "git remote add origin https://github.com/aryanrutheswar/map.git ... push it"        | Sync workspace to user's personal GitHub repository.                                                    | Configured remote origin, resolved untracked files, and pushed cleanly to <code>main</code> branch.                                                       |
| 06   | "This page doesn't exist 404 NOT_FOUND bom1:...."                                    | Vercel deployment was missing SPA static routing configurations.                                        | Created <code>vercel.json</code> rewrites and mirrored root assets to <code>web-app/public/</code> .                                                      |
| 07   | "show the temples and hotels and cafes and parks shopping malls tourist places"      | Map lacked categorized POIs and discovery features.                                                     | Engineered <b>POI Engine</b> : 6 categories, custom emoji pins, Explore Drawer, and Overpass API query integration.                                       |
| 08   | "when iam press upon a location 2 times it sould make a route to travels to travel"  | No driving navigation or travel simulation was present.                                                 | Implemented <b>Interactive 2-Press Travel Engine</b> : double-click Start/Destination, OSRM routing, glowing dual polylines, and animated car marker.     |
| 09   | "when iam in chitoor it should show the spots only in chitoor not in whole world..." | Previous pins were scattered worldwide (Paris, NY, Tokyo), leaking into local city view.                | Scoped spots strictly to <code>&lt; 30 km</code> of Chittoor, seeded 28 verified local landmarks, and implemented Haversine distance calculations.        |
| 10   | "y there is no option to selection the place... its only showing where i live"       | Geolocation was auto-running on page load; lack of dedicated Place Selection modal or map-click picker. | Removed automatic geolocation on load. Added <code>📍 Select Place</code> modal with quick cities and <b>interactive click-anywhere-on-map selection</b> . |

### 3. System Architecture & High-Level Design

The solution is structured as a modular frontend architecture paired with lightweight Node.js static hosting and live third-party geospatial microservices:



### 4. Directory Structure & File Roles

| PATH              | TYPE          | DESCRIPTION & CORE RESPONSIBILITIES                                                                           |
|-------------------|---------------|---------------------------------------------------------------------------------------------------------------|
| index.html        | HTML5         | Primary application structure: header, search bar, category chips, Explore Drawer, Telemetry HUD, and Modals. |
| style.css         | CSS3          | Custom design system: variables, glassmorphism, responsive drawers, custom pins, and route HUD styling.       |
| app.js            | JavaScript    | Complete application logic: Leaflet setup, POI engine, routing, geocoding, modal triggers, and JSX generator. |
| vercel.json       | JSON          | Vercel deployment route rewrites to ensure root static hosting without 404 errors.                            |
| web-app/server.js | Node.js       | Local development HTTP server serving web-app/public/ on port 3000.                                           |
| web-app/public/*  | Static Assets | Synchronized production mirrors of index.html , style.css , and app.js .                                      |

## 5. Points of Interest (POI) & Local Scoping Engine

A primary requirement was ensuring that when exploring a specific place (such as Chittoor), **no foreign or worldwide spots** appear. All displayed locations are strictly bound within a 25–30 km radius of the active focus area.

### Verified Chittoor Dataset (28 Local Landmarks)

| CATEGORY                                                                                    | PLACE NAME                                    | COORDINATES      | LOCALITY / NOTE                 |
|---------------------------------------------------------------------------------------------|-----------------------------------------------|------------------|---------------------------------|
|  Temple    | Kanipakam Sri Varasidhi Vinayaka Swamy Temple | 13.2989, 79.0357 | Historic Ganesha Temple (11 km) |
|  Temple    | Arthagiri Veeranjaneya Swamy Temple           | 13.3100, 79.0200 | Aragonda Hill Temple (13 km)    |
|  Temple    | Sri Venkateswara Temple Chittoor              | 13.2201, 79.1025 | Bazaar Street, Chittoor Centre  |
|  Temple    | Mogili Mugasura Someswara Swamy Temple        | 13.2833, 79.0833 | Mogili, Chittoor District       |
|  Hotel     | Bans The Hotel Chittoor                       | 13.2195, 79.1012 | Premium 3-Star Business Hotel   |
|  Hotel     | Hotel Bliss Chittoor                          | 13.2210, 79.1040 | Near Collectorate Office        |
|  Hotel     | Park Elite Hotel Chittoor                     | 13.2150, 79.0980 | Kongareddy Palli, Chittoor      |
|  Cafe    | Cafe Coffee Day Chittoor                      | 13.2205, 79.1030 | High Road, Chittoor Centre      |
|  Cafe    | Padmasri Cafe                                 | 13.0876, 79.0592 | Chittoor Highway Cafe           |
|  Park    | Gandhi Park Chittoor                          | 13.2165, 79.1005 | City Centre Public Park         |
|  Park    | Kaigal Waterfalls Nature Park                 | 13.0820, 78.6830 | Dumukurallu Waterfalls Reserve  |
|  Mall    | Reliance Mart Chittoor                        | 13.2047, 79.0971 | Supermarket & Shopping Centre   |
|  Mall    | CMR Shopping Mall                             | 13.2175, 79.1015 | Gandhi Road, Chittoor           |
|  Tourist | Chittoor Historic Clock Tower                 | 13.2170, 79.1008 | Heritage Clock Tower Square     |
|  Tourist | Gurramkonda Historic Hill Fort                | 13.7800, 78.5800 | 500-Year-Old Architectural Fort |
|  Tourist | Horsley Hills Hill Station                    | 13.6500, 78.4000 | Famous Scenic Hill Retreat      |

## 6. Interactive 2-Press Driving Route Engine

Users can double-click any two points on the map to calculate a live driving route and watch an animated vehicle drive along the road network.

### Navigation Workflow:

1. **Double-Click Location 1:** Drops Point A (● Start Marker).
2. **Double-Click Location 2:** Drops Point B (📍 Destination Marker).
3. **OSRM Routing:** Calls OSRM Driving API to fetch real roadway geometries.
4. **Polyline Rendering:** Renders a dual-layer polyline (cyan casing + sky blue core).
5. **Telemetry HUD:** Displays distance (km) and estimated drive time (mins).
6. **Animated Car:** A 🚗 vehicle marker moves along the route coordinates, rotating smoothly based on segment bearing angles.
7. **Controls:** Provides Pause/Resume, Replay, and 1x/2x/4x Speed toggles.

## 7. Place Selection Engine (No Home Auto-Lock)

Previously, the browser was automatically executing geolocation on load, which hijacked the map view and zoomed into where the user lives. This has been resolved with a dedicated multi-modal place selection system:

- **No Auto-Location Override:** Geolocation is strictly on-demand when the user clicks `My Location`.
- **Dedicated "Select Place" Button & Modal:** Located in both the top bar and category filter bar.
- **1-Click Quick Cities:** Direct access cards for Chittoor, Tirupati, Bengaluru, Chennai, Hyderabad, Mumbai, and Delhi.
- **Click Anywhere on Map to Select:** Tapping any location on the map drops an interactive pin, performs reverse geocoding, and shows a `📍 Show Nearby Spots` action button.

## 8. Code Walkthrough & Key Algorithms

### A. Haversine Distance Calculation (app.js)

```
function calculateDistanceKm(lat1, lon1, lat2, lon2) {
  const R = 6371; // Earth radius in km
  const dLat = (lat2 - lat1) * Math.PI / 180;
  const dLon = (lon2 - lon1) * Math.PI / 180;
  const a =
    Math.sin(dLat/2) * Math.sin(dLat/2) +
    Math.cos(lat1 * Math.PI / 180) * Math.cos(lat2 * Math.PI / 180) *
    Math.sin(dLon/2) * Math.sin(dLon/2);
  const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1-a));
  return R * c;
}
```

### B. Interactive Map-Click Place Selection (app.js)

```
map.on('click', async (e) => {
  if (activeTool === 'marker' || activeTool === 'geofence') return;

  const lat = e.latlng.lat;
  const lng = e.latlng.lng;

  if (!selectionMarker) {
    selectionMarker = L.marker([lat, lng], { icon: selectionPinIcon, zIndexOffset: 1200 }).addTo(map);
  } else {
    selectionMarker.setLatLng([lat, lng]);
  }

  const fullAddr = await reverseGeocode(lat, lng);
  const cityName = fullAddr.split(',')[0].trim() || 'Selected Location';

  const popupContent = document.createElement('div');
  popupContent.className = 'selected-place-popup';
  popupContent.innerHTML = `
    <div class="selected-place-badge"> 📍 Selected Location</div>
    <div class="selected-place-name">${escapeHTML(fullAddr)}</div>
    <button class="select-place-explore-btn"> 🔍 Show Nearby Spots in ${escapeHTML(cityName)}</button>
  `;

  popupContent.querySelector('.select-place-explore-btn').addEventListener('click', () => {
    focusOnArea(cityName, lat, lng, 14, 25);
    exploreDrawer.classList.add('open');
  });

  selectionMarker.bindPopup(popupContent).openPopup();
});
```

### C. 2-Press Driving Navigation Handler (app.js)

```
function handleLocationPress(latlng) {
  if (!routeStartMarker) {
    routeStartMarker = createRoutePointMarker(latlng, 'A');
    showToast('Origin set! Double-click destination to create route.');
```

```
} else if (!routeEndMarker) {
  routeEndMarker = createRoutePointMarker(latlng, 'B');
```

```
    calculateAndDrawRoute(routeStartMarker.getLatLng(), latlng);
  } else {
    clearTravelRoute(false);
    routeStartMarker = createRoutePointMarker(latlng, 'A');
    showToast('New start set! Double-click destination to create route.');
```

```
  }
}
```

```
map.on('dblclick', (e) => {
  handleLocationPress(e.latlng);
});
```

## 9. Bugs Diagnosed & Engineering Resolutions

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### 1. Worldwide Pin Leak into Local Views

*Problem:* Hardcoded pins for Paris, Tokyo, and New York were rendering even when the user moved to Chittoor.

*Solution:* Implemented geodetic radius filtering ( `dist <= radiusKm` ) inside `renderPOIs()` and replaced static mock pins with the verified Chittoor dataset.

### 2. Automatic Home Location Hijack

*Problem:* On page load, the browser immediately ran GPS geolocation and flew to the user's home.

*Solution:* Removed automatic `startLiveLocationTracking()` from script execution; geolocation is now strictly triggered on user click.

### 3. Horizontal World Duplication

*Problem:* Leaflet was wrapping the world horizontally, showing duplicate copies of continents.

*Solution:* Set `nowrap: true` on all tile layers and locked map bounds to `L.latLngBounds([-85, -180], [85, 180])`.

### 4. Vercel 404 Routing Error

*Problem:* Vercel was unable to resolve root static files.

*Solution:* Added `vercel.json` rewrites mapping root routes directly to `/index.html`.

## 10. User Testing & Verification Guide

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- Open Map:** Navigate to `http://localhost:3000`. Verify the map opens to a clean world view without auto-locating your home.
- Explore Chittoor:** Click  `Chittoor Spots` in the filter bar. The camera flies to Chittoor and displays 28 local landmarks.
- Filter Categories:** Click  `Temples` or  `Hotels`. Only Chittoor-area places are shown with real distance badges.
- Select Any Other Place:** Click  `Select Place` in the header, choose **Tirupati** or **Bengaluru**. Only spots in that city appear.
- Click on Map:** Click anywhere on the map. A pin drops with location details and a  `Show Nearby Spots` action.
- Plan a Route:** Double-click any two points on the map. Watch the OSRM route render and the animated car travel along the road.